

Template for Computer Programming I Practical Recording

Date	:	16/01/2025
Problem Specification	:	calculate the area for Square, Rectangle and Triangle
Assumption	:	The user input is valid integer.
Limitation	:	N/A
Input	:	Enter the width of the square, the width and length of the rectangle, and the width and height of the triangle
Processing	:	The area of the square, rectangle and triangle are calculated using Formula
Output	:	Display the area of the square, rectangle and triangle
Algorithm	:	Step 1: Prompt "Enter the width of the Square" Step 2: Initialize variable sqw for width. Step 3: Read the input sqw by user Step 4: Assign input value for sqw Step 5: Initialize areaOfSquare as $\text{areaOfSquare} \leftarrow \text{sqw} * \text{sqw}$ Step 6: Prompt "Enter the width of the Rectangle" Step 7: Initialize variable rew for width. Step 8: Read the input rew by user Step 9: Assign input value for rew Step 10: Prompt "Enter the length of the Rectangle" Step 11: Initialize variable rel for length. Step 12: Read the input rel by user Step 13: Assign input value for rel Step 14: Initialize areaOfRectangle as $\text{areaOfRectangle} \leftarrow \text{rew} * \text{rel}$ Step 15: Prompt "Enter the width of the Triangle" Step 16: Initialize variable triw for width. Step 17: Read the input triw by user Step 18: Assign input value for triw Step 19: Prompt "Enter the length of the Rectangle" Step 20: Initialize variable trih or length Step 21: Read the input trih by user Step 22: Assign input value for trih Step 23: Initialize areaOfTriangle as $\text{areaOfTriangle} \leftarrow \text{triw} * \text{trih} / 2$ Step 24: print value of areaOfSquare as "The Area Of Square is :" Step 25: Print value of areaOfRectangle as "The Area Of Rectangle is:" Step 26: Print value of areaOfTriangle as "The Area Of Triangle is :" Step 27: End the program

Programme listing	:	Program file attached
Test data and expected output	:	1. Test data: sqw=10,rew=18,rel=10.triw=10,trih=10 Expected output: The areaOfSquare is :100 The areaOfRectangle is :80 The areaOfTriangle is :50
Output obtained for test data	:	1. Test data: sqw=10,rew=8,rel=10.triw=10,trih=10 Expected output: The AreaOfSquare is :100 The AreaOfRectangle is :80 The AreaOfTriangle is :50
Discussion	:	Discuss the challenges or take away note for this program
Conclusion	:	Summarize the problem it aims to solve.